

A 25-year-old man who is wheelchair-bound presents for elective rigid cystoscopy. He had a C5 dislocation fracture with fusion done 1 year ago following a motor vehicle accident. He is able to shrug his shoulders and transfer himself from a seated position to the wheelchair.

A. What are the main anaesthetic concerns for this patient? (60%)

B. Discuss options of anaesthetic management for this patient. (40%)

A.

This is a young patient presenting for elective urological procedure, possibly for evaluation of neurologic bladder secondary to chronic spinal cord injury (SCI) at C5 level.

Patient

Cardio

Autonomic dysreflexia (ADR)

- Clinical emergency characterized by unopposed sympathetic response secondary to stimulus below level of lesion, is a potential risk during surgical manipulation.
- A history of flushing/ sweating/ headache during urinary catheterization would suggest high risk for ADR.
- I will insert arterial line for closer monitoring, and adjust my anaesthesia/ analgesia as appropriate.
- Rescue medications (remifentanyl/ fentanyl, beta-blockers like esmolol) should be readily available for treatment of ADR.

Increased risk of arrhythmias

- Secondary to vagal hypersensitivity.
- Any concerning symptoms (palpitation/ chest pain) or ECG features would warrant further work up by cardiology.

Increased risk of cardiovascular disease

- Due to decreased physical activity.
- I would screen for angina symptoms and perform pre-op 12-lead ECG for strain pattern/ ischemia changes.

Decreased circulating blood volume

- Which means that patient may not handle drop in SVR/ preload well.
- Patient may benefit from slow, titrated induction of GA, and IV fluid loading pre-induction/ CNB.

Airway/ Respi

Airway instrumentation may be challenging.

- Because of previous cervical spine instrumentation limiting neck extension.
- Should GA be required, low threshold for awake fiberoptic intubation (AFOI) if there are other features suggestive of difficult airway.

Increased risk of respiratory infections/ peri-operative respiratory complications.

- Due to impaired cough/ poor clearance of secretions.
- Op should be postponed if active respiratory tract infection present.
- Extended post-op monitoring in PACU may be required to monitor for respiratory complication (atelectasis) post-mechanical ventilation.

GI

Gastroparesis

- May be present, which increases aspiration risks, and may necessitate RSI intubation.

Neuro

Extrajunctional ACh receptors

- Predispose patient to hyperkalaemia if depolarising NMBA such as Suxamethonium is used. This should be avoided. Rocuronium may be used if RSI is required.

Higher risk of psychiatric disorders (adjustment disorders/ depressions).

- Necessitating careful checks for drug-drug interactions (SSRI-Tramadol/ Pethidine).
- Certain psychotropic medications may be sedative.
- Use of illicit drugs/ drug abuse should also be asked during history taking.

May have chronic pain syndrome as a result of SCI

- Pain history should be elicited, with additional care taken as appropriate.
- Baseline pain medications should be checked to avoid double dosing/ facilitate titration of other analgesics.

Thermoregulation is often impaired

- Necessitating intra-op temperature monitoring and prompt correction

Renal

Neurogenic bladder

- Predisposes to renal injury/ impairment.
- To avoid worsening renal impairment peri-operatively, ensure adequate preload/ maintain renal perfusion pressure and avoid nephrotoxic drugs.
- Drug doses may need to be adjusted.
- Frequent urinary catheterization predisposes patient to latex allergy, which require additional preventive measures in OT.

Wheelchair bound. Complications associated with prolonged immobility include:

Elevated venous thrombo-embolic (VTE) risk

- Consideration for mechanical VTE prophylaxis
- Patient may be on anti-coagulant, which would need to be stopped pre-op for surgical/ central neuraxial block.

Contractures/ spasticity/ pressure sores are common.

- And should be checked pre-op to facilitate padding at bony prominences/ pressure sore areas.
- Careful position to minimise injury during lithotomy position.

Potential difficult IV access

- Due to skin atrophy/ dependent oedema.
- US would be useful in difficult IV access situations.

Anaesthesia

Patient may not require much anaesthesia/ analgesia for the procedure.

- In complete spinal cord injury with absent sensation below level of lesion
- Focus will be on (1) prevention and management of ADR and (2) ensuring patient's comfort.

Spinal anaesthesia is effective in prevention of ADR but positioning for central neuraxial block may be challenging.

- If patient is unable to sit up unsupported (lateral position may be required).
- Difficult to ascertain level of sensory block if sensation is impaired.

Surgical

Instrumentation/ dilation/ stretching of ureters may be stimulating and trigger ADR.

- Close communication with surgeon and close monitoring during stimulating phases will facilitate prompt interventions should ADR occurs.

B.

Options of anaesthesia management include:

1. Monitored anaesthesia care (especially if there is complete loss of sensation below level of lesion)
2. Regional anaesthesia (spinal)
3. General anaesthesia

1. Monitored anaesthesia care (especially if there is complete loss of sensation below level of lesion)

- This would be least invasive in terms of airway instrumentation and procedures (spinal).
- However, management of ADR would be the most challenging.
- Firstly, the amount of sedation/ analgesia doses in MAC may not be sufficient to prevent ADR.
- My ability to increasing sedation/ analgesia is limited and has to be balanced against ability to maintain airway and ventilation.
- If patient has underlying gastroparesis, sedation increases risk of aspiration.

2. Regional anaesthesia (spinal)

- A working spinal anaesthesia is effective in prevention of ADR.
- It is also superior in terms of peri-op pain control, and avoid complications associated with airway instrumentation, mechanical ventilation, and anaesthesia/ RSI drugs.
- However, positioning for central neuraxial block may be challenging if patient is unable to sit up unsupported (lateral position may be required).
- Difficult to ascertain level of sensory block if sensation is impaired.

3. General anaesthesia

- I would perform a gastric US pre-induction to confirm empty antrum.
- Assuming good mouth opening and no other features of difficult airway, I would then proceed with use of supraglottic airway (SGA) with NGT in-situ to decompress stomach at the end of the operation.
- Ultra-short acting opioids remifentanyl for rapid titration in response to urinary/ureteric instrumentation (including dilatation/ stretch/ stenting) to avoid complications associated with long acting opioids. Paracetamol/ tramadol may be given for post-op discomfort.
- I would also request for surgeon to administer lignocaine gel prior to scope insertion to minimise surgical stimulus.
- I would also have b-blockers such as esmolol/ labetalol readily available in OT in response to ADR.
- This avoids the need for laryngoscopy in a patient with limited neck extension because of previous cervical spine instrumentation while maintaining control of airway/ ventilation. It also allows me to focus on haemodynamics management intra-operatively by adjusting depth of anaesthesia, analgesia and use of anti-hypertensive medications.
- It will be more challenging if patient has difficult airway. Low threshold for AFOI, or RSI intubation using Rocuronium.
- Otherwise, this would be my preferred option as it confers me the most control over patient's airway and haemodynamic management.